

# CURRICULUM SCIENTIFIC AND EDUCATIONAL

Of

ANTONELLA GUZZO

(drafted pursuant to and for the purposes of articles 46, 47 and 76 of Presidential Decree 445/2000)

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## 1 PERSONAL INFORMATION

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## 2 POSITIONS HELD

### 2.1 CURRENT POSITION

- **Associate University Professor** for the scientific-disciplinary sector ING-INF/05-Information processing systems at the Department of Computer Engineering, Modeling, Electronics and Systems (DIMES) of the University of Calabria since 02/04/2018.
- From 5 May 2021, with Rectoral Decree No. 692 of 2021, **Member of the University Quality Assurance Committee (PQA)**. The PQA is a University body that oversees the adequate and uniform implementation of Quality Assurance (QA) processes and procedures – at University level, in Departments and Degree Courses – on the basis of current legislation and the guidelines formulated by the Governing Bodies. As a member of the PQA working group, he/she participates in the activities of promoting quality within the University, in the organization and coordination of data collection activities for monitoring indicators, ensuring that the timing and information flows are managed consistently with the QA model developed by the University.
- From 25 April 2021, **Department Quality Representative** appointed by the Director of DIMES. As a representative, he/she provides training and information support to all the actors involved in the quality process at the departmental level and at the level of the individual degree courses, directs the interested actors towards specific training and information initiatives on the topics of the AQ and on the contents of the PQA Guidelines, monitors the timing, role and tasks of the responsible actors. In particular, he/she coordinates the execution of the processes, monitors them, providing the appropriate indications in case of misalignments. Participates in the drafting of the departmental strategic plan, in the hearings of the Evaluation Unit and in the activities of the AQ. He/she detects critical issues during the execution of the processes and supports the responsible actors in defining the possible corrective actions to overcome them.
- He is currently a member of the University commission for the drafting of the Quality guidelines.
- Referent for the Industry 4.0 Curriculum of the national doctorate in DLT and BLOCKCHAIN which has its main headquarters at the University of Camerino [P hD\_DLT&BK].
- Since 2022 (first edition), member of the National Doctoral College in DLT [P hD\_DLT&BK] and contact person appointed by the university for the college itself.
- Since 2008, member of the ICT Doctoral College of DIMES.

- From 2021 to July 2023, president of the evaluation commission appointed by the ICT doctoral college of DIMES to express an opinion on the transition of doctoral students to the years following the first and express an opinion for the final evaluation commission.
- Since 2018, member of the teaching commission of the Degree Course in Management Engineering at the University of Calabria.
- Member of the following associations: National Interuniversity Consortium for Computer Science (CINI), Cy-bersecurity, IEEE, ACM, AIXIA and IEEE Women.
- Member of the AIXIA working group in ARTIFICIAL INTELLIGENCE FOR HEALTH. (<https://aixia.it/gruppi/hc/>).
- Member of the Italian Distributed Ledger Technology Working Group, part of CINI. (<https://dltgroup.dmi.unipg.it/index.php>)
- Member of the CINI Digital Health Laboratory at the University of Calabria.
- Pertaining to the Cybersecurity National Lab. of CINI.
- Coordinator of the local unit (of the University of Calabria) for the European project Behavioral Next Gen-eration in Wireless Networks for Cyber Security (BEiNG-WISE) Cost Action CA22104, and representative for Italy of the project Management Committee (<https://www.cost.eu/actions/CA22104/#tabs+Name:Management%20Committee>).
- Co-Principal Investigator of a PRIN project (2022 call) Sec.7.1)
- Founding member since 2018 and member of the Board of Directors of the start-up and university spin-off EVO-BI.
- Partner of the university spin-off OKT srl of the Relatech group.
- Responsible for WP 9.6 within the FAIR project (Section 7.1)
- Spoke 9 contact for KPIs within the FAIR project (Section 7.1)
- Spoke 9 contact for the Data Management Group of the FAIR project
- Head of OS3 of the Radioamica project (Section 7.1)
- Head of OR1 of the SOLIDO project (Section 7.1)

## 2.2 POSITIONS HELD IN THE PAST

|                          |                                                                                                                                                                                                                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2007-2008:               | Research Associate at the Institute for High Performance Computing and Networking (ICAR-CNR), with provision prot. n. 667/2007.                                                                                                                                                             |
| from<br>2014 to 12/2019: | With DR n.1191 of 26/6/2014 and with DR n.974 of 29/07/2016 he covered the role and functions of President of the "Direction and Control Commission" established to carry out direction and control functions of the activities of the childhood hub service of the University of Calabria. |
| from<br>2013 to 12/2019: | With DR n.2599 of 10/12/2013 she covered the role and functions of "Rector's Delegate for the coordination and supervision of the correct performance of nursery school activities and with DR n.1687 of 10/09/2014 of the service of the Childhood Center of the University of Calabria.   |
| from 07/27/2005          |                                                                                                                                                                                                                                                                                             |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| as of 11/27/2005:                    | He held a collaborative position with the Department of Electronics, Computer Science and Systems (DEIS) of UNICAL in Rende (CS), within the Miur-FAR project called "Intelligent and Adaptive Environments to Support e-Government and e-Democracy in Local Public Administrations".                                                                                                                                                      |
| from 01/04/2004<br>to 30/09/2004:    | He held a collaborative position with the Institute for High Performance Computing and Networks (ICAR-CNR) in Rende (CS), within the project "Prototype creation of a system for the definition of a value-added service chain".                                                                                                                                                                                                           |
| from 02/09/2003<br>as of 03/01/2004: | He held a collaborative position with the Institute for High Performance Computing and Networks (ICAR-CNR) in Rende (CS), within the project "Prototype creation of a system for the definition of a value-added service chain".                                                                                                                                                                                                           |
| from 03/23/2002<br>as of 11/30/2002: | He held a research position on behalf of the Institute for High Performance Computing and Networks (ICAR-CNR) in Rende (CS), within the project "Evolution of a Multi-Channel Front Office System (for virtual channels) for on-line Banks".                                                                                                                                                                                               |
| from 04/09/2001<br>as of 28/02/2002: | He held a consultancy role for the Calabria Region in the context of the "Regional Information Society Initiative (RISI)", project "Arianna - Second Round", approved by the European Commission.                                                                                                                                                                                                                                          |
| from 01/10/2000<br>to 30/09/2001:    | He held a research position on behalf of the Institute for Systems and Computer Science (ISI-CNR) of Rende (CS), within the project "Data Warehouse for the Calabria Telematic Plan (DW-PTC)" for the analysis and re-engineering of the information flows of the institution and the preparation of feasibility studies in order to foresee new systems to support the statistical service and strategic planning of the Calabria Region. |
| from 12/28/1999<br>to 09/26/2000:    | He held a scientific collaboration position with the Institute for Systems and Computer Science (ISI-CNR) of Rende (CS), within the project "Development of Experimental Space-Time Modeling of Environmental Evolutionary Processes for Risk Mitigation".                                                                                                                                                                                 |

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### 3 TRAINING ACTIVITIES

#### 3.1 UNIVERSITY STUDY

April 11, 2000 She obtained a degree in Management Engineering at the University of Calabria, discussing the thesis "Proposal for an algorithm for the economic and safe dispatching of thermoelectric power: in the preventive phase" and obtaining a grade of 110/110 with honors. The thesis supervisor was Prof. Daniele Menniti. In her thesis work she studied and developed an algorithm for the search for an optimal solution to the problem of "Economic and safe dispatching of thermoelectric power for an electric power system", in collaboration with Prof. Jacques Carpentier (International consultant for Power Systems for Electricité de France- EDF)

26/06/2000 He passed the state exam to qualify as an engineer at the University of Calabria.

### 3.2 POST -GRADUATE TRAINING

from 1/11/2001                      He attended the PhD course in Systems Engineering and Computer Science at the Department of Electronics,  
to 31/10/2004                      Computer Science and Systems (DEIS) of the University of Calabria.

During the course, he attended various seminars organized both at the University of Calabria and at other universities, and participated in international schools reserved for PhD students.

12/02/2005                      He obtained the title of Doctor of Research in Systems Engineering and Computer Science, at the University of Calabria, defending the thesis "Languages, Algorithms and Methods for Modeling, Reasoning on, and Mining Workflows". The thesis supervisor was Prof. Dominic Saccà.

### 3.3 SCHOLARSHIPS AND RESEARCH GRANTS

from 1/11/2001                      She was awarded a 3-year scholarship by the University of Calabria to attend the PhD course in Systems Engineering  
to 31/10/2004                      and Computer Science.

in 2005                              She was awarded a research grant lasting 18 months at the Institute for High Performance Computing and Networking (ICAR-CNR) in Rende (CS), within the European project called "INFOMIX Boosting the Information Integration".

in 2006                              She was awarded a 3-year research grant for young researchers at the Institute for High Performance Computing and Networks (ICAR-CNR) in Rende (CS), as part of the project funded by the Ministry of University and Research within the FIRB program (Fund for Investments in Basic Research) called "TOCAI.it Knowledge-Oriented Technologies for Business Aggregations for the Internet".

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## 4 NATIONAL SCIENTIFIC QUALIFICATION

In January 2020 he obtained the national scientific qualification as Full Professor for the competitive sector H1/09, Scientific Disciplinary Sector ING-INF/05 with unanimous approval of the commission.

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## 5 RESEARCH ACTIVITIES

The research activity is part of the representation and discovery of knowledge from data analysis, touching over the years on both purely theoretical aspects [see e.g. [ 19, 20, 21, 24, 26]] and applicative and technological ones (see e.g. [8, 10, 12, 15, 35]), the latter in line with the implementation of research projects funded by the ministry because they are considered to have a high impact on the growth and competitiveness processes of the Italian and international system. For over 20 years he has been involved in Process Mining, carrying out research and dissemination of scientific results, promoting technology transfer and collaborating with national and international research groups in dissemination actions (see activity carried out as a member of the IEEE task force, IEEE working group, Italian Group on Process Mining). She was the proponent of the establishment of a university course in Process Mining at the University of Calabria for the master's degree in Management Engineering and the master's degree in Computer Science.

Since the approval date of the proposal, she has been a full professor for the course in Management Engineering and co-professor, as can be seen from the course syllabus, in Computer Science. In recent years, she has focused her research activity on distributed systems and in particular on the development of Federated Learning applications and traceability with Blockchain.

Research on Federated Learning covers three important topics: (i) the development of distributed applications for

diagnosis, prognosis and the study of the oncological therapeutic response through medical imaging (see the activity carried out as OS3 manager for the Radioamica project); (ii) the development of green-aware algorithms and systems (see the activity carried out as co-PI and Pilot manager of WP6 in Spoke 9 of the FAIR project); and (iii) the analysis and study of the impact of No-IID data (not identically and uniformly distributed) in federated learning (see the activity of PRIN 22 of which he is co-proposer and unit manager). As regards the research activity on blockchain, for several years he has been working on projects for traceability of goods, proposing cooperation models between companies that use blockchain for their traceability processes in the supply chain. She joined the Italian DLT (Distributed Layer Technologies) group and, in the year in which the national doctorates were established, she was called to be part of the DIN national doctoral college of Camerino and referent of the Industry 4.0 Curriculum of the doctoral program.

In his research activities, he has also addressed issues related to inverse frequent mining, workflowgraph mining and coclustering. The main research lines and the results obtained are detailed below:

## 5.1 PROCESS MINING

In this context, attention has been paid not only to traditional Process Mining techniques, but also to the extension of such techniques with innovative approaches, which exploit Data Mining techniques for the analysis of process execution logs. Process management systems are systems aimed at supporting their execution in different application contexts [15, 17], through appropriate mechanisms for specifying and monitoring activities and execution constraints. They are increasingly used in the automation of production processes, from traditional industries to companies specialized in software production, and of management processes. Classical issues related to process automation are related to the design and optimization of the execution scheme [30], on which the expenditure of available resources crucially depends. For example, a good planning of a logistics process for sorting goods in a port area can lead to significant reductions in maneuver times (and costs), thus opening the space for an increase in sustainable traffic.

In order to support the (re-)design and optimization phase of processes, we have mainly dealt with the definition of process mining techniques, thanks to which the most promising activity scheduling strategies are identified on the basis of past choices and recorded in the execution logs. Differently from the traditional approach, in which a descriptive model of the process executions is extracted from the log, we have proposed a technique that has proved to be pioneering in the field of process mining, in which an algorithm is proposed to cluster process logs, i.e. to partition the various executions based on appropriate similarity criteria, in order to reconstruct their use cases and typical variants. In particular, in [35] we have addressed the problem of structural clustering, where executions are discriminated on the basis of the activities involved, while in [34] the technique has been extended to consider the data associated with the activities, such as execution time or executors. These techniques have been supported by mechanisms for consolidating and interpreting the acquired knowledge [28,32].

Still in the context of process management, in [46] a technique is proposed to identify anomalous process instances outliers that can be expressions of exceptional situations such as malfunctions, pathologies, and anomalies in general. The discovery of such instances constitutes an important source of information in various situations such as, for example, in the identification of frauds, in the identification of intrusions in computer systems, in personalized marketing, in medical workflows, and in data cleaning. In particular, in [29] the discovery of outliers is based on the induction of decision trees starting from the execution logs, to be used at run-time to identify possible anomalous situations and estimate the evolution dynamics of the executions themselves.

A further line of research has been the definition of a process discovery technique in which the model learning uses a set of constraints on the typology of the graph underlying the model, able to express a sort of “background knowledge” of the process. In [26], a computational analysis of the problem is presented, and a prototype system that implements the constraint programming technique for the induction of the model.

## 5.2 FEDERATED LEARNING

Federated learning [12] is an emerging distributed computational paradigm in which, differently from the traditional approach, the nodes participating in federated learning share the partial results of the learned models (parameters, gradient, local model), without sharing the data. In this context, the research activity concerns three specific aspects: 1) study and definition of new federated learning models that guarantee adequate levels of accuracy even in the case of NO-IID [4] (not identically distributed) datasets; 2) study and definition of new green-by-design federated learning models, i.e. that can reduce aggregation rounds and energy consumption [1, 3, 5] and 3) the development of distributed applications for diagnosis, prognosis and the study of the oncological therapeutic response through medical imaging [12]. One of the most successful application areas of federated learning is radiomics. Radiomics is a quantitative approach to the analysis of medical images (e.g., acquired by computed tomography, magnetic resonance imaging, or positron emission tomography) in combination with molecular, genetic, and clinical information. The research activity consists in proposing new approaches based on Artificial Intelligence tools and methods that are more effective than traditional diagnostic/prognostic approaches and are able to accurately characterize a series of multifaceted physiopathological processes, ranging from oncological to cardiac and other diseases.

Furthermore, these approaches will be supported by XAI explainability techniques, aimed at identifying the factors that most influence the decision-making process of artificial intelligence and are able to support learning outcomes [14,2].

## 5.3 BLOCKCHAIN

Blockchain and Distributed Ledger (DLT) are explicitly mentioned in the most relevant strategic planning documents of our country and Europe (e.g.: PNR 21-27, PNRR 21-26 and related complementary funds). The research activity consists in defining methods and frameworks to support the safe, secure and responsible identification of entities and actions operating on the blockchain with particular reference to scenarios involving business units and consortia. In addition to identity management, the research includes the development of new mechanisms and protocols that add a security layer on DLT (Distributed Ledger Technology), blockchain and smart contracts [ 10, 11, 13].

## 5.4 DEEP LEARNING MODELS FOR TIME-BASED DATA ANALYSIS AND APPLICATION DEVELOPMENT

### SMART

The research activity has focused on the study and development of deep learning architectures for the analysis of temporal data. In particular, different types of deep models have been used, including convolutional networks, generative models and architectures for time series. The following generative approaches have been proposed: 1- VAE (Variational AutoEncoder). A framework and a deep architecture for the generation of synthetic data based on user-described features have been proposed. The applicability of the technique in the context of image generation has been demonstrated and the results have been included in the publication .

I have proposed a completely innovative approach that exploits generative VAEs for the automatic generation of medical reports from radiographic images [5]. 2- (Marked) Temporal Point Process -TPP [16] . TPPs are an elegant mathematical framework used to model the generative process of a temporal sequence of events. Many real phenomena produce data that can be represented as temporal events (events related to earthquakes, tornadoes and more generally to objects that move according to a trajectory). Behind these apparently random events there are complex mechanisms, for example earthquakes cause new earthquakes in the form of aftershocks and an essential tool to model these mechanisms is a stochastic process that also allows to make predictions not only on the type of future event, but also on the time of occurrence. Much research has been based on parametric (or semi-parametric) assumptions on the intensity function based on 1) the previous knowledge of the processes, and 2) the maximum likelihood estimation (MLE) that is used



to adapt the model parameters from the data. These models work well when the parametric assumptions are correct, but, in many cases where the generative process of the real event is unknown, these parametric assumptions may be too limited and not reflect reality. In this context, the use of deep learning architectures has been proposed to learn the distribution and intensity function of events, thus being able to capture the non-linearity and complexity of the relationships between events even very distant from each other in the past history. The approach has been used and validated in the context of Human Activity Recognition and for intrusion detection in IDS systems [7]. Furthermore, deep architectures have been used for the development of smart applications in the context of Human Activity Recognition [8] and Healthcare [9].

## 5.5 CO-CLUSTERING TECHNIQUES AND THEIR APPLICATIONS

The problem of simultaneously clustering objects of heterogeneous typologies has been widely analyzed in recent years, due to its significant impact in several application domains. High-order co-clustering consists in simultaneously partitioning multiple heterogeneous domains based on correlation data between values of different domains, typically represented in the form of contingency tables. Most of the proposals in the literature are limited to the case of only two domains (bi-clustering) and, in the very few approaches designed for a greater number of domains, the problem is addressed by minimizing a linear combination of the objective functions related to individual bi-clustering sub-problems. This requires that each contingency table be associated with a weight, which expresses the reliability and relevance of the data contained in it for the purposes of co-clustering. Currently, one of the open problems in the literature is the definition of automatic methods for tuning the weights, since it is known that their choice is not a simple task and strongly affects the results of co-clustering. In light of this issue, the work in [49] was presented, where an information-theoretic framework for the high-order co-clustering problem is introduced, in the case of correlations between domains structured according to a “star”, with a central domain correlated with all the remaining domains. Rather than optimizing a single global objective function (with some explicit weighting scheme), an appropriate notion of agreement is introduced between the various two-dimensional co-clustering subproblems, which guarantees a minimum level of approximation of the (local) optimal solution for each single objective function. In the same work, an algorithm for the efficient computation of a co-clustering, which follows this notion of agreement, was defined. This approach was applied to the analysis of workflow process logs, under the assumption that they contain data related to process performance measures in [34]. The basic idea is to formalize a co-clustering problem, where the trace domain is related to various numerical domains that each correspond to a particular performance metric.

In this scenario the various domains are clustered simultaneously, based on an information-theoretic framework, extended with the introduction of appropriate techniques for the partitioning of numerical domains. Finally, in [43] an extension of the coclustering algorithm is provided in a parallelized version, using the popular Map-Reduce paradigm.

Motivated by the difficulty of appropriately selecting such weights in the objective function, in this context a technique has been studied that automatically and implicitly defines the various weights, efficiently finding the combination that best optimizes the overall performance [31].

## 5.6 INVERSE FREQUENT SET MINING

The inverse frequent set mining problem consists in finding (if it exists) a database in which a specified set of itemsets are frequent. Recently, this problem has become very sensitive in the scientific community (and not only in the Data Mining community) due to its implications in the context of privacy preserving data mining and in the definition of benchmark data. Specifically, given the reluctance of owners of sensitive data (e.g. process logs, web purchase logs, etc.) to make such logs available for analysis or to test mining algorithms, inverse frequent mining is proposed as a method for generating synthetic datasets from a set of mining results (frequent itemsets). Most of the scientific contributions have focused on the computational aspects, highlighting the intractability of the problem in its most general formulation, which is based on the definition of constraints expressed by itemsets and their frequency. The contribution of the research activity

[44] was to define a more general framework, where supports are considered instead of frequencies and the constraints on the itemsets are specified with an upper bound and lower bound of the support instead of an exact value. The general problem and some variants were then studied by imposing restrictions on the generation of the synthetic dataset. In addition to the complexity results, which allowed us to define a complete and detailed picture of the computational analysis of the main problem and of some cases of particular interest, we then worked on defining and proposing a heuristic for solving the problem itself. In [27], the original formulation of the inverse frequent set mining problem was extended by placing limits also on infrequent itemsets. This formulation, if on the one hand allowed us to define more realistic application scenarios, on the other hand inevitably worsened the intrinsic complexity conditions of the problem. The research conducted in [27] identified two alternative approaches to the problem, (i) defining a parametric formulation so as to identify those conditions in which the problem instance becomes tractable, (ii) providing an approximate solution obtained by applying the column generation technique, as a variation of the simplex technique that is more efficient in solving problems formulated in linear programming with an exponential number of variables.

## 5.7 GRAPH MINING TECHNIQUES APPLIED TO WORKFLOWS: WORKFLOW MINING

The use of workflow systems as a tool for analyzing and monitoring processes in different application domains has created a strong interest in the study of techniques for mining execution logs. A first line of research has been focused on mining instances or parts of them frequently scheduled in a system. The techniques developed in this area aim at (i) run-time, to support the workflow user in the decision-making process, analysis and understanding of the behavior; (ii) design-time, to identify anomalous situations attributable to an incorrect design of the schemes. In the first case, the logs actually recorded by the system during the execution of the workflow are used, with the aim of identifying critical activities (bottlenecks, fictitious tasks), of identifying discriminating activities that characterize the success/failure of an execution and in general of identifying which choices in the past have more frequently led to the achievement of a desired final configuration. In the second case, the instances generated by the simulation are used to identify deadlock conditions, critical situations and evaluate, for example, the executability conditions of a node.

In the framework of this research, which represents a completely new and unprecedented approach to workflow mining, several algorithms and related specializations have been presented in [36,33]. These algorithms are based on a formalization of the workflows and instances scheme, which has allowed to formally prove the intractability of some typical problems in this domain, motivating the use of data mining techniques. In particular, techniques have been proposed for the identification of unexpected correlations between sub-processes, which can be used to characterize discriminant activities and to identify Patterns of interest.

Other mining techniques have been defined for the identification of “frequent patterns”, i.e. workflow substructures frequently scheduled by the system, whose knowledge aims (at run-time) to support the workflow user in the decision-making process of analyzing and understanding the behavior, and (at design-time) to identify anomalous situations attributable to an incorrect design of the schemes [36].

## 5.8 USING ATTACK PATTERNS FOR INTRUSION DETECTION

He is currently studying a new line of research in the field of computer security [73] and more precisely in Intrusion detection where the intrusion represents an improper use of the system, a real cyber attack, etc. and everything that identifies actions that violate the rules imposed by security. In [67] a hypergraph model with various expressed constraints (temporal and cardinality constraints) is proposed, descriptive of various types of attack. The idea is to use this model to identify the sequence of activity logs that constitute an intrusion. To this end, in addition to the formal framework, the computational complexity of the consistency analysis of the logs to the model and the discovery of anomalous instances has been characterized. In [25] the analysis of the hypergraph model is completed with a theoretical analysis of equivalence and minimality and an algorithm is presented, based on the appropriate definition of an index and related algorithms for its maintenance and recovery, capable of efficiently identifying attacks by analyzing instances of the model.

## 5.9 COALITION STRUCTURE GENERATION

In chronological order, the last line of research on which he has focused his attention concerns the development of clustering techniques specifically for use in multi-agent systems contexts. The goal is to define appropriate logical formalizations of the characteristics and capabilities of agents, to then formally reason on how such agents, in a cooperative context, can organize themselves in coalitions to obtain greater benefits than they could obtain if they acted in isolation [40]. In particular, he has focused his attention on contexts in which agents are modeled in terms of marginal contribution networks (a simple propositional formalism well known in the field of cooperative game theory) and in which domain constraints can be expressed on the coalitions that can be formed. In this context, he has analyzed the computational complexity of the clustering problem and has identified tractable cases identified on the basis of the topological properties of the underlying networks [24].

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## 6 ORGANIZATION, MANAGEMENT AND COORDINATION OF NATIONAL RESEARCH GROUPS AND INTERNATIONAL, OR PARTICIPATION IN THE SAME

The main experiences gained in the management and participation in research groups (national and international) and in the conduct of studies entrusted by qualified public or private institutions are reported below.

### 6.1 MANAGEMENT OF A GROUP OF RESEARCHERS

He is currently responsible and coordinates (under his scientific supervision and responsibility) the research activities of the following components:  
SECURITY

- Michele Ianni. She is the scientific director of the RTDA position currently held by Dr. Ianni (since 1-1-2022) on the topic "Cybersecurity - Advanced methods and techniques for security and privacy in emerging systems and applications for IoT". She was also the scientific director of the research grant awarded from 2018 to 2019 on ""Innovative methodologies and technologies for the design and development of infrastructures for the accountability of cooperative services, also based on blockchain infrastructures".

#### BLOCKCHAIN GROUP

- Fiza Fiyal. He is the supervisor of the doctoral student, winner of the selection for the 38th cycle in the doctorate in ICT of Dimes, and the scientific director of the doctoral scholarship assigned to him.
- Usman Tahir. He is the supervisor of the PhD student Tahir, winner of the 38th cycle selection and the scientific PhD in DLT and blockchain - curriculum in Industry 4.0 - PhD awarded, director of the scholarship to him.

#### Deep/FEDERATED LEARNING

- Asaf Raza. He is co-supervisor of the PhD student PhD in ICT from Dimes, and scientific director of the won the selection at the 38th cycle in I scholarship awarded to him.
- Dipanwita Thakur. She is the scientific director of the RTDA position currently held by Dr. Thakur, winner of the FAIR 7.1 call selection
- MUHAMMAD IKRAM FARWA IKRAM. He is co-supervisor of the PhD student in ICT at DIMES, winner of the selection of the FAIR 7.1 call

## 6.2 PARTICIPATION IN NATIONAL AND INTERNATIONAL RESEARCH GROUPS

Since 2021 he has been collaborating with the Mathematical modelling and Data Analysis research group (MODAL) of the University of Naples Federico II. Specifically, together with the group he is developing new methodologies, applications and services for federated learning (see PRIN 2022 sect.7.1).

Since 2018 he is a senior member of the SPEME (Smart, PErvasive and Mobile systems Engineering) research group of the University of Calabria. Specifically, he is responsible for research activities related to the topics of Deep/Machine Learning and Federated Learning.

From 2005 to 2018 she collaborated steadily with the "Architecture of Information Systems (AIS)" group led by Prof. Wil van der Aalst, of the Technical University of Eindhoven, Netherlands, on process mining issues, where she was a guest having covered the role of Visiting Researcher.

From 2003 to 2020 she collaborated permanently with the Institute for High Performance Computing and Networks (ICAR-CNR), of which she was formally a research associate with provision prot. n. 667/2007, in 2007 and 2008 to provide her contribution within the Strategic Commissions: 1) commission ICT.P08.002 - Managing, Extracting Knowledge and Reasoning on Data, 2) commission INT.P01.010 - ESTRADA: Reliability and Advanced Data Analysis and Security Support.

In 2013 he collaborated with the group led by Prof. D. Cotroneo, of the Department of Electrical Engineering and Information Technologies, University of Naples Federico II, on topics related to Anomaly and Intrusion detection.

## 6.3 SCIENTIFIC STUDIES AND RESEARCH ENTRUSTED BY QUALIFIED PUBLIC OR PRIVATE INSTITUTIONS

During her association with the CNR, from 2007 to 2008 she was responsible for a study commissioned by the Institute for High Performance Computing and Networking of the National Research Center (ICAR-CNR) on "Innovative techniques for data analysis and management" within two projects: 1) CT.P08.002 - Managing, Extracting Knowledge and Reasoning on Data, 2) project INT.P01.010 - ESTRADA: Reliability and Advanced Data Analysis and Security Support.

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## 7 RESEARCH PROJECTS

### 7.1 PARTICIPATION IN PROJECTS WITH RESPONSIBILITY ROLES

**Project Manager/Director:** He is the scientific manager of a research contract awarded by the Relatech SpA group to the Department of Computer Engineering, Modeling, Electronics and Systems of the University of Calabria, within the Industrial Research and Experimental Development project: The Digital Transformation of Brands in the Fashion Sector: New Emerging Technologies for the Management of the Production and Distribution Network (Digital Brand), CUP: B49J23001940005, Call issued by the Ministry of Business and Made in Italy (hereinafter MIMIT) for the application sectors of the National Strategy for Intelligent Specialization relating to "Intelligent Factory", "Agrifood" and "Life Sciences". In the contract, he assumed scientific responsibility for the implementation of research activities on the following implementation objectives: OR 3 – RI "Certification models in the Fashion Supply Chain based on Distributed Ledger Technology (DLT)", • OR 6 – RI "Machine Learning (ML) algorithms, techniques and models in the Fashion Supply Chain", • OR 8 - SS "Traceability platform based on DLT and ML models applied to the Supply Chain in Fashion".

**CO-Principal Investigator:** He is responsible for WP 9.6 called "AI for GREEN" of SPOKE 9: GREEN-AWARE AI of the project "Future Artificial Intelligence Research (FAIR)" funded under the National Recovery and Resilience Plan, Creation of Enlarged Partnerships

extended to Universities, Research Centres, Enterprises and funding basic research projects to be funded under the National Recovery and Resilience Plan (NRRP), Mission 4 Component 2 Investment 1.3 funded from the European Union – NextGenerationEU, with Decree Directorial No. 1243 of 02/08/2022. Specifically, WP9.6 defines AI solutions to enable sustainable growth in different application domains, with particular attention to applications that have a broad impact on citizens' lives, and is divided into 5 tasks in which four different research groups belonging to four different research institutes collaborate for their expertise as many scientific disciplinary sectors. Spoke 9 of the project includes a total of six WPs and it is organized so as to have two referents for each WP who are responsible for the general activities planned and must also coordinate the groups working on the WP activities.

**HUB - FAIR Referent:** As part of his participation in the FAIR project, he plays the role of Referent for Spoke 9 of the KPI activity. He is also part of the Data Management Group defined at Hub level for the management of the project Data Management Plan.

**Pilot Manager:** He is responsible for the pilot "AI for distributed IoT applications" within WP 9.6 of the FAIR Project sect.7.1. This pilot project aims to define a framework for a "green" application in the context of the Internet of Things, using Federated Learning for EdgeAI as a New Green-by-Design Computing and Communication Paradigm for Distributed Smart Devices (Wearables and Smartphones, tablet). The WP is in turn divided into 5 tasks.

**OS leader:** He is responsible for the OS3 workpackage: "Secure & Privacy-Preserving AI and XAI for the cooperative construction of models" of the RADIOAMICA project – Open Network for "Cooperative RADIOmics/rAdiogenoMICs based on Artificial Intelligence" – funded within the Development and Cohesion Plan Health, Trajectory 2 "E-Health, advanced diagnostics, medical devices and minimally invasiveness". The project is divided into 10 workpackages and involves 3 universities, 2 IRCSS, 1 National Research Institute and 4 Hospital Companies. In the specifically the OS3 for which he is responsible consists of three tasks for analysis, design, and the implementation of AI and Explainable AI algorithms and models that learn patterns predictive diagnostics in a federated and privacy-preserving manner.

**OR 1 leader:** He is responsible for the implementation objective OR1 within the scope of project consultancy SOLID (Efficient solutions for the design and management of a Distribution Center) that the DIMES and DIMEG departments have stipulated with PAC logiastic 2000. In particular, OR1 called: Business Intelligence Methodologies and Tools for Processes Logistics is divided into 3 Industrial Research sub-tasks

**PRIN 2022:** She is the winner as secondary proposer of the PRIN2022 called GANDALF - Gan Approaches for Non-iiD Aiding Learning in Federation. The GANDALF project aims to address one of the major challenges of Federated Learning (FL), the non-IID problem, in light of the broader scenario of Edge Artificial Intelligence (Edge-AI), integrating innovative machine/deep learning methodologies (i.e. GAN-based approaches).

**Project Manager/Director:** She was the scientific manager of a research project entrusted by the ICT-SUD Competence Center to the Department of Computer Engineering, Modeling, Electronics and Systems of the University of Calabria, within the Research Project ARS01\_00587 "SON - DISTRIBUTED LEDGERS FOR SECURE OPEN COMMUNITIES", admitted financed by Concession Decree no. 3272 of 06-12-2018 by the Ministry of Education, University and Research (MIUR), project code n° ARS01\_00587. ICT-SUD's consultancy in favour of DIMES-UNICAL concerns the responsibility on OR 1 (Evolutionary Scenarios, Business Models and Integration Platforms); OR 4 (Distributed

Ledger for the sharing economy); OR 5 (Distributed Ledger for copyright); OR 7 (Pro-totyping and Experimentation of Distributed Ledger for the sharing economy) and OR 8 (Pro-totyping and Experimentation of Distributed Ledger for copyright). Project start 1-1- 2019 Project end 31-12-2021 ICT-SUD consultancy in favor of DIMES-UNICAL for a total amount of 140,000, of which 118,000 for Industrial Research activities and 28,000 for Experimental Development activities.

Project Manager/Director: She was the scientific manager of a research contract awarded by the ICT-SUD Competence Center to the Department of Computer, Modeling, Electronics and Systems Engineering of the University of Calabria, within the project F/080007/03/X35 "Protect ID: Innovative processes and technologies for the protection of digital identities and personal information online", CUP B28I17000500008, PON-MISE Grandi Progetti di R&S Call for Proposals of June 2016. The duration of the research contract is from March 17, 2017 to December 31, 2019. The project is carried out by ICT-SUD together with four other industrial partners: Engineer-ing SpA (coordinator), Poste Italiane SpA, Alkemy SpA, Resiltech srl. Universities and research centers do not participate directly in the project since the call for proposals provides for their involvement only as consultants, to be explicitly declared in the project proposal, as occurred in the assignment of the research contract to the University of Calabria by ICT-SUD. The contract provides for the implementation of research activities on the implementation objectives: OR7 "Management of large amounts of data relating to authenticated transactions between users and service providers"; OR8 "Intelligent system for security assessment and risk rating of service providers"; OR 12 "Prototype of an intelligent system for security assessment and risk rating of service providers based on innovative big data management techniques". The total cost recognized for the research contract activities is Euro 200,000 (net of VAT), of which Euro 90,000 for OR7, Euro 55,000 for OR8 and Euro 55,000 for OR12.

Unit Manager/Director: He is the head of the research unit of the Department of Computer Science, Modeling, Electronics and Systems of the University of Calabria for the project "Digital Identity and Service Accountability" (IDService) for the implementation objective "OR1: Models and techniques for the use of digital identity in cooperative service scenarios", of which he coordinates and directs the activities and collaborates with the other industrial partners of the project: OKT srl and H2I srl. The purpose of the project, funded under the PON -MISE 2014-2020 with an industrial entity as lead partner, is to provide innovative solutions in the field of decentralized service provision in which different actors, potentially belonging to different and independent organizations and domains, need to base their mutual interaction on "strong" trust relationships. from 01-01-2016 to 31-12-2019.

Scientific Director: He is the Scientific Director of n. 1 research grant relating to the Digital Identity and Service Accountability Research Program (IDService) at the Department of Computer Engineering, Modeling, Electronics and Systems (DIMES) - UNICAL, issued with DD n° 8 of 10/01/2018, pursuant to Law n. 240/2010, articles 18, 22, 24 and 29 and the Regulation for the Award of Grants for collaboration in research activities, approved with DR n. 1655 of 08/09/2014.

Scientific Manager: Scientific Manager of the three-year research contract, signed on 29 ÿ 1 ÿ 2018 between the company Fidens srl and the DIMES department. The contract is aimed at designing an integrated system for controlling and reporting business data, including the study and design of connectors/gateways to existing heterogeneous platforms and the development of the basic elements for a new programming language related to Business Intelligence. It also includes the definition of algorithms capable of supporting business choices based on the available integrated data.

Scientific Director: He is the Scientific Director of n. 1 research grant awarded, through selection publishes among 43 grants put out to competition with university funds, to the project entitled "CYBER BALANCE - Cyber-security risks in the era of technologically-mediated crime: balancing security, protection, privacy, and costs", DR 931 of 07/19/2017 and DR 108 of 01/25/2018.

Scientific Director: Since 2013 he has been responsible for ex 60% projects, within research units of the Department of Computer Engineering, Modelling, Electronics and Systems of the University of Calabria.

## 7.2 PARTICIPATION IN PROJECTS AS A MEMBER OF THE WORKING GROUP

Tech4You Technologies for climate change adaptation and quality of life improvement", Piano National Recovery and Resilience, Mission 4, Component 2, Investment 1.5.

Cyber Security: The Cyber Security Technology District is part of the development and strengthening initiatives of the Ministry of Education, University and Research (MIUR) relating to "Districts High Technology, public/private laboratories as well as actions aimed at creating new Districts and/or new Public/Private Aggregations" within the National Operational Programme "Research and Competitiveness" 2007-2013 (PON R&C) for the Convergence Regions (Calabria, Campania, Puglia, Sicily). 3 research projects are planned with prevalent industrial research activities (about 70%) and the remaining part of experimental development, each lasting about 2 years (until December 2015). It is currently included as a member of the research unit of the University of Calabria in the project "Safe Dematerialization (Project Code: PON03PE\_00032\_3)". The general objectives of the project are the definition of Reference models and process specification for safety control in the various life cycle phases of digital documents and the creation of an end-to-end platform for the secure management of the life cycle of a digital document, which includes adequate tools for the coverage of individual security checks.

FRAME: Since May 2013, as a member of the research unit of the University of Calabria within the project "FRAME: Flexible and expandable FRAMework, based on "Methodologies and tools based on knowledge, for the consolidation and optimized management of complex information systems", project PON0102477/F4, Operational Programme National Research and Competitiveness 2007-2013), financed by the MIUR from the European Fund for Regional Development (ERDF) and the Revolving Fund (RFR) for the implementation of the Policies Community. The project aims to create a framework to support and make the process of using applications and corporate information bases, even unstructured and multimedia, efficient in a service composition logic. The partners involved are Siemens/Altos (leader), Exeura Srl, the University of Calabria, the Institute of High Performance Computing and Networks (ICAR-CNR). The project lasts 3 years.

PRIN TENACE: Since September 2012, as an external unit member associated with the research unit of Reggio Calabria, he participates in the project "TENACE: Protection of National Critical Infrastructures from Cyber Threats", funded by the Ministry of Education, University and Research. Research, within the "PRIN 2012" projects. Main objective of the TENACE project is the definition of organizational methodologies and collaborative techniques that increase the protection of ICed in particular financial infrastructures, power grids and systems transport, from cyber threats following, where possible, a collaborative approach. The project lasted 3 years.

PON TETRIs: Since January 2011, as a member of the research unit of the University of Calabria within the project "TETRIs: Innovative Open Source Services on TETRA". The project is

has been approved by the Ministry of Education, University and Research on the National Operational Programme Research and Competitiveness 2007-2013 for the convergence regions (project PON0100451), and is co-financed by the European Regional Development Fund (ERDF) and the Revolving Fund (FDR) for the implementation of Community Policies. The research unit to which it belongs involves professors and researchers from the University of Calabria (Department of Mathematics), the University of Pisa (Department of Computer Science), the University "Ca' Foscari" of Venice (Department of Computer Science), and the University "La Sapienza" of Rome (Department of Computer Science and Systems). The project partners are "Orangee Srl", "Selex Elsag Srl" "ICT-SUD", "ICAR-CNR", "H2i srl", "Exeura Srl", "Sinap-sys Srl", "TSC Srl", "CNIT", the University of Calabria, and the University of Reggio Calabria. The aim of the project is to create a framework that allows the construction of innovative technological components that exploit the potential of both the TETRA network and the corresponding commercial networks to create complex and articulated Smart Environments, intended as fundamental elements of and constituting a Smart Country/Smart City, aimed at providing advanced services such as emergency management, environmental control, and mobility in urban areas. The project lasted 3 years.

OpenKnowTech: Since July 2007, as a member of the research unit of the University of Calabria of the project "OpenKnowTech: Laboratory of Technologies for the Integration, Management and Distribution of Data, Processes and Knowledge" within the Realization Objective "Open Source for the management of processes and Business Process Intelligence". The project aims at the creation of a public-private laboratory of open-source software technologies, was approved by the MIUR with Ministerial Decree (DM) 2630/Ric of 30 November 2006, and was co-financed by the European Regional Development Fund (ERDF) of the European Union.

Partners of the project are—in addition to the University of Calabria—the Institute for High Performance Computing and Networking of the National Research Council (ICAR-CNR), Exeura srl, HERZUM Software Solutions Center srl, ID Technology srl, and Orangee srl. The purpose of the realization objective is the definition and implementation of systems for the automated management of processes with advanced reasoning functionalities to support the modeling, monitoring and management phases of process executions. The laboratory is located at the ICT-SUD Competence Center, Piazza Vermicelli, Polo Tecnologico UNICAL, 87036 Rende (CS). The laboratory has completed the start-up phase, and will have an operation phase of at least another 5 years. She has been specifically involved in the research activities of the OR6 realisation objective which essentially include: (i) the analysis, design and development of systems and environments for the management of business processes, equipped with tools for the analysis of process models, in comparison with the real behaviour found in practice; (ii) the definition and testing of solutions for Process Mining, capable of discovering frequent patterns and process models, detecting any anomalies and supporting the optimisation of future executions.

TOCAI.IT: From July 2006 to April 2010, he participated as a referent for industrial research activities in the TOCAI.IT project: knowledge-oriented technologies for business aggregations on the Internet, co-financed by the Ministry of University and Research within the FIRB program (Fund for Investments in Basic Research). The aim of the project was to develop an integrated set of methodologies, techniques and software tools based on the most advanced knowledge technologies, which allow the analysis, specification, implementation and evaluation in the field of new business organization models, in an "internetworked enterprise" perspective. To this end, the project envisaged the substantial involvement of 3 companies representing 3 different organizational models of the innovative enterprise: intra-company integration, inter-company integration at the supply chain level, and cooperation at the district level. The challenge of the project is to identify a set



coherent combination of advanced technological methodologies and solutions suitable for the three cases, overcoming the limits and fragmentation of products and services currently available for the integrated company, and above all contributing to creating a corporate culture more aware of the potential and tools of knowledge technologies. In the context of this project, he contributed as a member of "activity 9" to the development of Knowledge Discovery techniques for the discovery and classification of intra- and inter-organizational processes and knowledge, aimed at understanding the ways in which cooperation is carried out and knowledge is shared.

AUTOMATON:

Since July 2007, as a member of the research unit of the University of Calabria in the project "AUTOMA (AUTomobile logistic MANagement)", funded by POR CALABRIA 2000-2006. The project had as its general objective the development of a support system for the management and optimization of logistic processes concerning the distribution of cars within a maritime hub, and embraced the following industrial research problems: (i) tracking and localization of cars through radio frequency systems; (ii) implementation of adaptive, multi-channel and multimodal interfaces; (iv) optimization methods for loading and unloading operations; (v) advanced techniques for workflow management to support the control, analysis and optimization of logistic processes. In particular, she dealt with the activities related to the implementation objectives OR.3 ("Workflow management for the provision of advanced services and support for the distribution of automobiles"), OR.5 ("Architecture of the AUTOMA system and methodology for the integration of modules") and OR.6 ("Support methodologies for the definition and development of integrated logistics processes"), with particular reference to the study of methods and techniques for the analysis of the execution of logistics processes.

INFOMIX:

From September 2003 to March 2005, he was a researcher at the Institute for High Performance Computing and Networking of the National Research Council (ICAR-CNR) within the project INFOMIX: boosting information integration, co-financed by the European Community under the Fifth Framework Programme (IST-2001-33570). The project was promoted by a consortium formed by the University of Calabria (Italy), the University of Rome "La Sapienza" (Italy), the "Vienna University of Technology" (Austria), and the company "Rodan Systems" (Poland). The goal was to realize an innovative information integration system in the following points: (i) data reasoning functionality, (ii) semistructured data management. In the project, he contributed both to the basic research and to the actual design of the system.

G4B:

From September 2, 2003 to September 2005, he received two research assignments from the Institute for High Performance Computing and Networking of the National Research Council (ICAR-CNR) within the project: "Prototype realization of a System for the definition of a value-added service chain (called G4B)". The general objective of the project was the realization of an intelligent design system based on Workflow design as an enabling technology for the realization of service chains. This system allows the modeling of the service workflow through the integration and composition of services (basic and non-basic) guided by the results obtained by querying a "Knowledge Base" (for example, the ontological descriptors of services and the functionalities offered). In the project, he worked on the definition and implementation of algorithms specialized in the mining of workflow instances to support the design and monitoring of processes. He also coordinated the implementation of a prototype that supports these functionalities.

MIUR - FAR:

From 27 July 2005 to 27 September 2005 and from 1 January 2006 to 1 March 2006 he received two collaboration assignments with the Department of Electronics, Computer Science and Systems.

(DEIS) of UNICAL of Rende (CS), within the Miur project n.54 called “Intelligent and Adaptive Environments to Support e-Government and e-Democracy in Local Public Administrations”. The project had as its general objective the conception, modeling and prototyping of a system to provide high value-added services to Citizens and Businesses (e-democracy) and to Administrators and Officials of PALs (e-government). Within the scope of the above-mentioned project, in the first phase it dealt with the identification and analysis of a set of administrative procedures of interest in relation to the possible adoption of workflow tools for defining administrative processes and procedures. He also conducted an analysis of the methodologies, based on innovative mining techniques execution logs, for monitoring and improving process management and on interoperability between systems through the use of web services. In the second phase of the project, he worked on the development of a demonstrative software for application cooperation and process integration between different PALs using Web Services and on the experimentation of the system on data coming from the Taxes, Registry and Aqueduct information subsystems of the municipality of Catanzaro.

ex 60%: From 2007 to 2013 he participated continuously in ex 60% projects, within units of research of the Department of Computer Engineering, Modeling, Electronics and Systems of the University of Calabria.

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## 8 PROGRAM COMMITTEES, EDITORIAL COMMITTEES AND REVIEW ACTIVITIES

### 8.1 PRESIDENT OF PROGRAM COMMITTEES

- He is workshop chair of the workshops presented at the 23rd edition of the international conference: International Conference of the Italian Association for Artificial Intelligence (AIxIA 2024)
- General chair of the seventh, eighth and ninth editions of the workshop in Process Querying, Manipulation, and Intelligence (PQMI2022, PQMI2023 and PQMI2024) at the leading international conference in Process Mining, International Conference on Process Mining (ICPM2022, ICPM2023 and ICPM2024)
- Session Organizers of the Special Session on Federated Learning on Big Data for the international conference IEEE Big Data 2024
- Session Chair of the IoT&Security workshop co-located with the 21st IEEE International Conference on Dependable, Autonomic & Secure Computing (DASC 2023)
- General chair and signatory of the proposal of the workshop in Process Management in the AI era (PMAI 22@IJCAI) at the 31st edition of the leading international conference in Artificial Intelligence, INTERNATIONAL JOINT CONFERENCE ON ARTIFICIAL INTELLIGENCE IJCAI
- Session chair of the workshop in Process Management in the AI era (PMAI 22@IJCAI) at the 31st edition of the leading international conference in Artificial Intelligence, INTERNATIONAL JOINT CONFERENCE ON ARTIFICIAL INTELLIGENCE IJCAI.
- General chair of the workshop in Process Management in the AI era (PMAI 23@IJCAI) at the 32nd edition of the leading international conference in Artificial Intelligence, INTERNATIONAL JOINT CONFERENCE ON ARTIFICIAL INTELLIGENCE IJCAI
- Program co-chairs of the conference FIFTH INTERNATIONAL CONFERENCE ON BLOCKCHAIN COMPUTING AND APPLICATIONS (BCCA 2023)

- General Chair for the 2nd edition of IoT Security workshop at the 21st IEEE International Conference on Dependable, Autonomic & Secure Computing, DASC 2023 - Chair, IoT Security workshop at 20th IEEE International Conference on Dependable, Autonomic & Secure Computing, DASC 2022
- Publication Chair for the conference The International Conference on Embedded Wireless Systems and Networks (EWSN 2023)
- Session Chair for the session – Security and Trust at The International Conference on Embedded Wireless Systems and Networks (EWSN 2023).
- Organizer of the PhD workshop for 1st International School on Internet of Things & Edge AI: Computing, Communications and Systems, Sept 8- 12, 2022, Falerna (CS), Calabria, Italy, sponsored by the IEEE SMC Chapter of the IEEE Italy Section .
- Local chair and member of the organizing committee of the “29th edition of the Italian conference on Information Systems Advanced for Databases (SEBD21)
- Publicity and coordination chair, and member of the organizing committee of the “Workshop on Business Process Intelligence (BPI'06)”, held in conjunction with the “International Conference on Business Process Management (BPM'06), Vienna, Austria, 4-8 September, 2006.
- Member of the organizing committee of the Join Conference on Declarative Programming (APPIA- GULP- PRODE 2003).
- Member of the organizing committee of the eleventh “Italian Symposium on Advanced Database Systems (SEBD'03)”, Cetraro (CS), Italy, June 24-27, 2003.

## 8.2 MEMBER OF PROGRAM COMMITTEES

She has been or is a member of the program committee of the following international conferences (or workshops):

- “The 21st IEEE International Conference on Dependable, Autonomic & Secure Computing DASC 2023”  
Nov 14-17, 2023
- “The 5nd International Conference on Process Mining ICPM 2023, Rome, Italy, October 23-27 2023.
- the 21th International Conference on Business Process Management BPM2023 Netherlands, September 11-15, 2023
- The International Conference on Embedded Wireless Systems and Networks (EWSN 2023) , September, Rende, Italy.
- 1st AlxIA Workshop on Artificial Intelligence For Healthcare (HC@AlxIA 2022), Udine, Italy, November 30, 2022
- “The 20st IEEE International Conference on Dependable, Autonomic & Secure Computing DASC 2022”  
Calabria, Italy, Nov, 2022
- Workshop on Digital Twins for Business Processes, co-located with The 20th IEEE International Conference on Pervasive Intelligence and Computing (PICom 2022)
- Workshop on COMMunity-OrieNted WEARrable Computing Systems, colocated with The 7th IEEE Cyber Science and Technology Congress September 12-15, 2022
- “The 4nd International Conference on Process Mining ICPM 2022, Bolzen, Italy, October 23-28 2022.

- the 20th International Conference on Business Process Management BPM2022 Munich, Germany, September 11-16, 2022
- The 30th International Joint Conference on Artificial Intelligence (IJCAI-21)
- ISDA 2021: 21st International Conference on Intelligent Systems Design and Applications, Seattle, WA, United States, December 13-15, 2021
- the 19th International Conference on Business Process Management BPM2021 Rome, Italy, September 6-10, 2021
- AIOT4DH 2021: 1st Workshop on Artificial Intelligence and Internet of Things for Digital Health, Washington DC, DC, United States, December 16-17, 2021
- "The 3rd International Conference on Process Mining ICPM 2021, Eindhoven, Netherlands, October 31-November 5, 2021
- T-ECMLPKDD-2021 (Journal Track for the 2021 European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases)
- "Thirty-Fourth AAAI Conference on Artificial Intelligence AAAI'20", New York, USA, February 7-12, 2020.
- "The 1st Workshop on Blockchain and Enterprise Systems BES 2020", Riga, Latvia, November 25-27, 2020.
- "The 2nd International Conference on Process Mining ICPM 2020, Padua, Italy, October 6-8, 2020.
- "International Joint Conferences on Artificial Intelligence Organization (IJCAI'20)", Yokohama, Japan, July 11-17, 2020
- XXIV "International Database Engineering & Applications Symposium (IDEAS'20)", Seoul, South Korea, June 3 - 5, 2020.
- XXVIII "Italian Symposium on Advanced Database Systems (SEBD'20)", June 21-24, 2020 - Villasimius, Sardinia, Italy
- ML4PM2020: First International Workshop on Leveraging Machine Learning in Process Mining, Padua, Italy, October 5, 2020
- "International Joint Conferences on Artificial Intelligence Organization (IJCAI'19)", Macao, PR China, August 10-16, 2019
- XXIII "International Database Engineering & Applications Symposium (IDEAS'19)" Harokopio University, Athens, Greece, June 10-12, 2019
- ICPM "International Conference on Process Mining (ICPM'19)", Aachen/Germany, June 24-26, 2019.
- XXII "International Database Engineering & Applications Symposium (IDEAS'18)", Villa San Giovanni, Italy, June 18-20, 2018
- XIV "Workshop on Business Process Intelligence (BPI'18)", held in conjunction with the "International Conference on Business Process Management (BPM'18)", Sydney, Australia, September 9 - 14, 2018
- XXIX "International Conference on Database and Expert Systems Applications (DEXA'18)", Regensburg, Germany, September 3 - 6, 2018.

- XXVI<sup>y</sup> "Italian Symposium on Advanced Database Systems (SEBD'18)", Castellana Grotte (Taranto), ITALY, 24–27 JUNE, 2018
- VIII<sup>y</sup> "IEEE Symposium on Computational Intelligence and Data Mining (CIDM'17)", Hawaii, Honolulu, November 27 - December 1, 2017.
- XXVIII<sup>y</sup> "International Conference on Database and Expert Systems Applications (DEXA'17)", Lyon, France August 28 - 31, 2017.
- XIII<sup>y</sup> "Workshop on Business Process Intelligence (BPI'17)", held in conjunction with the "International Conference on Business Process Management (BPM'17)", Barcelona, Spain. September 10-15, 2017
- VI<sup>y</sup> "IEEE Symposium on Computational Intelligence and Data Mining (CIDM'16)", Athens, Greece December 6-9.
- XXVII<sup>y</sup> "International Conference on Database and Expert Systems Applications (DEXA'16)", Porto, Portugal September 5 - 8, 2016.
- XII<sup>y</sup> "Workshop on Business Process Intelligence (BPI'16)", held in conjunction with the "International Conference on Business Process Management (BPM'16)", Rio de Janeiro, Brazil September 18-22, 2016.
- V<sup>y</sup> "IEEE Symposium on Computational Intelligence and Data Mining (CIDM'15)", Cape Town, South Africa, December 7-10, 2015.
- XXVI<sup>y</sup> "International Conference on Database and Expert Systems Applications (DEXA'15)", Valencia, Spain September 1 - 4, 2015.
- XI<sup>y</sup> "Workshop on Business Process Intelligence (BPI'14)", held in conjunction with the "International Conference on Business Process Management (BPM'15)", Innsbruck, Austria, August 31 - September 3, 2015
- V<sup>y</sup> "IEEE Symposium on Computational Intelligence and Data Mining (CIDM'14)", Orlando, Florida 9-12 December, 2014.
- V<sup>y</sup> "IEEE Symposium Series on Computational Intelligence (SSCI 2014)", Orlando, Florida, USA, 9-12 December, 2014
- XXV<sup>y</sup> "International Conference on Database and Expert Systems Applications (DEXA'14)", Munich, Germany, September 1-5, 2014.
- VI<sup>y</sup> "the International Joint Conference on Knowledge Discovery and Information Retrieval (KDIR'14)", part of the IC3K conferences, Rome Italy, October 21-24, 2014.
- X<sup>y</sup> "Workshop on Business Process Intelligence (BPI'14)", held in conjunction with the "International Conference on Business Process Management (BPM'14)", Haifa, Israel, 7-11 September, 2014.
- XXIV<sup>y</sup> "International Conference on Database and Expert Systems Applications (DEXA'13)", Prague, Czech Republic, August 26-30, 2013.
- V<sup>y</sup> "the International Joint Conference on Knowledge Discovery and Information Retrieval (KDIR'13)", which is part of the IC3K conferences, Vilamoura, Spain, September 19-22, 2013.
- IX<sup>y</sup> "Workshop on Business Process Intelligence (BPI'13)", held in conjunction with the "International Conference on Business Process Management (BPM'13)", Beijing, China, August 26-30, 2013.

- XXIII<sup>y</sup> "International Conference on Database and Expert Systems Applications (DEXA'12)", Vienna, Austria, 3-6 September, 2012.
- IV<sup>y</sup> "the International Joint Conference on Knowledge Discovery and Information Retrieval (KDIR'12)", which is part of the IC3K conferences, Barcelona, Spain, October 4-7, 2013.
- VIII<sup>y</sup> "Workshop on Business Process Intelligence (BPI'12)", held in conjunction with the "International Conference on Business Process Management (BPM'12)", Tallinn, Estonia, September 3-6, 2012.
- XXII<sup>y</sup> "International Conference on Database and Expert Systems Applications (DEXA'11)", Toulouse, France, August 29-September 2, 2011.
- III<sup>y</sup> "the International Joint Conference on Knowledge Discovery and Information Retrieval (KDIR'11)", which is part of the IC3K conferences, Paris, France, October 26-29, 2011.
- VII<sup>y</sup> "Workshop on Business Process Intelligence (BPI'11)", held in conjunction with the "International Conference on Business Process Management (BPM'11)", Clermont-Ferrand, France, August 28- September 2, 2011.
- XXI<sup>y</sup> "International Conference on Database and Expert Systems Applications (DEXA'10)", Bilbao, Spain, August 30-September 3, 2010.
- II<sup>y</sup> "International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management (KDIR'10)", Valencia, Spain, October 25-28, 2010.
- VI<sup>y</sup> "Workshop on Business Process Intelligence (BPI'10)", held in conjunction with the "International Conference on Business Process Management (BPM'10)", Hoboken, NJ, USA, September 13-16, 2010.
- XX<sup>y</sup> "International Conference on Database and Expert Systems Applications (DEXA'09)", Linz, Austria, August 31-September 4, 2009.
- V<sup>y</sup> "Workshop on Business Process Intelligence (BPI'09)", held in conjunction with the "International Conference on Business Process Management (BPM'09)", Ulm, Germany, 7-10 September, 2009.
- IV<sup>y</sup> "Workshop on Business Process Intelligence (BPI'08)", held in conjunction with the "International Conference on Business Process Management (BPM'08)", Milan, Italy, September 1-4, 2008.

### 8.3 EDITORIAL COMMITTEES

- Editor of IoT Springer Book "Activity Recognition and Prediction for the smart environments based on the "Internet of Things" 2024
- Guest Editor of the Special Issue "Exploring the Frontiers of Radiomics: Unveiling Novel Insights through Advanced Techniques and Multimodal Approaches" for the journal Computer Methods and Programs in Biomedicine, Elsevier.
- Guest Editor of the Special Issue "Data Fusion for Federated Learning" for the Information Fusion journal, Elsevier.
- Guest Editor of the Special Issue in "Federated Learning on the Edge: Challenges and Future Directions" for the journal Future Generation Computer Systems.
- Since August 2022 he has been on the Editorial Board of the journal "Discover Artificial Intelligence", Springer.
- Since November 2022 he has been on the Editorial Board of the journal "Engineering Applications of Artificial Intelligence" (EAAI), Elsevier .

- Guest Editor (2021) in "Process Mining and Emerging Applications", a special issue in Algorithms-MDPI journal.

#### 8.4 OTHER ACTIVITIES AS A REVIEWER OF SCIENTIFIC PUBLICATIONS

He also participated in the review of various scientific articles proposed for publication in journals, national and international conferences and workshops, such as:

- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML - PKDD22)
- The 19th IEEE International Conference on Ubiquitous Intelligence and Computing (UIC 2022)
- The Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI-21)
- The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI-20)
- Siam International Conference on Data Mining (SDM),
- International Conference on Database Theory (ICDT) (2003, 2005, and 2012),
- International Conference on Database and Expert Systems Applications DEXA (2009 and 2011),
- The 15th European Conference on Logics in Artificial Intelligence (JELIA 2016) European Conference on Principles and Practice of Knowledge Discovery PKDD (2005),
- International Conference of Business Process Management BPM (2016, 2015, 2012 and 2005),
- International Symposium on Information Technology ITCC (2004),
- Joint Conference on Declarative Programming (APPIA-GULP-PRODE),
- ACM Symposium on Applied Computing SAC (2003),
- IEEE Symposium Series on Computational Intelligence SSCI (2013).
- IEEE Transaction on Knowledge and Data Engineering (TKDE)
- Expert Systems With Applications;
- ACM Transactions on Knowledge Discovery from Data (TKDD);
- Review Board for the Special Issue of the Journal on Intelligent Information Systems (Springer, <https://www.springer.com/journal/10844>) 2022.
- Special Issue on Knowledge Representation and Reasoning for Healthcare Processes
- Data Mining and Knowledge Discovery, Springer;
- Artificial Intelligence In Medicine
- Simulation Modelling, Practice and Theory (Elsevier)
- Data and Knowledge Engineering (DKE, Elsevier),
- Knowledge and Information Systems (KAIS), Springer
- Pervasive and Mobile Computing,

- AI Communication
- Information Systems (IS, Elsevier),
- ToPNoC Special Issue on Concurrency in Process-Aware Information Systems (Lecture Notes in Computer Science), Journal of Intelligent Information Systems (Springer)
- Oxford University Press (The Computer Journal), IET Software (IET Digital library).

## 8.5 ACTIVITY OF REVIEWER OF RESEARCH PROJECTS

|      |                                                                                                                                                                                                                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | She was appointed as an expert for the evaluation of projects presented to apply to the ERDF Notice no. 01/2023 "Support for the development of innovative start-ups in the areas of intelligent specialization of Trentino. Specifically, 3 projects were assigned by the Industry, Research and Mining Service Industry and Research Office of the Province Autonomous of Trento |
| 2023 | She has been appointed for the year 2023 as a reviewer for research projects funded by the National Fund for Scientific and Technological Development, Government's FONDECYT PROGRAM of Chile (Ministerio de Education, Gobierno de Chile). Specifically, it revised n°1 project (proposal N°1230697) for the current year.                                                        |
| 2014 | As a process mining expert and auditor for the Ministry of Education, he reviewed of the University and Research (MIUR), a project funded by the MIUR, within the SIR (Scientific Independence of young Researchers) program.                                                                                                                                                      |
| 2012 | He reviewed, as a process mining expert, a project funded by the Austrian Science Fund (FWF), as part of a program of Austrian academic research activities.                                                                                                                                                                                                                       |
| 2010 | He reviewed, as a process mining expert, a project in the SSD ING-INF/05, within the framework of a call for proposals formulated by the University of Verona for the implementation of joint projects with local institutions and companies.                                                                                                                                      |
| 2007 | He reviewed, as a process mining expert, a project funded by the Dutch Science Research Council within a Dutch research activity program.                                                                                                                                                                                                                                          |

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## 9 ORGANIZATION AND COORDINATION OF OTHER SCIENTIFIC ACTIVITIES

- He is a member of the Scientific Committee of the PNRR Advanced Management Training Course - M6C2 - 2.2.3 – "Development of technical-professional, digital and managerial skills of healthcare system personnel - Sub-measure: managerial training course".  
<https://sites.google.com/unical.it/corsoaltaformazione/home>
- He is an organizing member of the CyberChallenge.IT for the University of Calabria headquarters <https://cyberchallenge.it/venue/unical> in the years 2019, 2020, 2021, 2022, 2023, 2024. Cyber-Challenge.IT is the first Italian cybersecurity training program for young people in schools higher education and universities. The program is organized annually by the National Cyber-security Laboratory of the National Interuniversity Consortium for Computer Science (CINI) with the support of the Cyber-security System Information for the Security of the Italian Republic. Now in its third edition, CyberChallenge.IT today it has 18 universities.



- He is a founding member of the IEEE task force in Process Mining (the only Italian researcher together with Prof. Alessandro Sperduti). The task force was established in 2009, within the Data Mining Technical Committee (DMTC) of the Computational Intelligence Society (CIS) of the Institute of Electrical and Electronic Engineers, Inc. (IEEE), and is made up of international experts who aim to promote and disseminate research, development and training in Process Mining.
- She was the organizer and chair of the "Process Mining Dissertation Award", a prize for the best thesis of in the context of Process Mining (2014-2015-2016 edition).
- It is a founding member of the IEEE Working Group, a group of international experts who worked on the definition of the XES Standard - eXtensible Event Stream - to ensure interoperability with event logs that record processes in monitoring systems (e.g., ERP and workflow), also in the context of event streams. The standard passed the approval process by the IEEE Computational Intelligence Society in November 2016 and is officially available in the IEEE Digital Library (<http://ieeexplore.ieee.org/document/7740858/>, DOI 10.1109/IEEESTD.2016.7740858). The approval process of the document was concluded in July 2017 ([http://www.win.tue.nl/ieetfpm/doku.php?id=news:2017\\_07\\_20\\_xes](http://www.win.tue.nl/ieetfpm/doku.php?id=news:2017_07_20_xes)).
- She was a member of the Scientific Council of the 2nd level Master's Degree in "Information Security" academic year 2017/2018 which will take place at the University of Calabria starting from November 2017. The Master's Degree, which has obtained the involvement of national and international partners, including Poste Italiane, NTTdata, Global Cyber Security Center, aims to create professional figures capable of: defining and governing the main processes, policies and organizational models of information security; correctly managing risks and estimating the potential effects of attacks on technological infrastructures and information assets and identifying the most suitable security measures; ensuring compliance with the main regulations and reference standards for the protection of personal data and corporate information assets.

2021 He is a member of the technical and organizational committee of the IEEE Technical Committee on Hyper-Intelligence (<https://hyper-intelligence.org/index>)

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## 10 RELEVANCE AND VISIBILITY OF SCIENTIFIC ACTIVITY

She is among the signatories of the document drawn up by 25 international experts that defines the XES Standard - eXtensible Event Stream - for the interoperability of event logs that record processes in monitoring systems (e.g., ERP and workflow), also in the context of event streams. The standard passed the approval process by the IEEE Computational Intelligence Society in November 2016 and is officially available in the IEEE Digital Library (<http://ieeexplore.ieee.org/document/7740858/>, DOI 10.1109/IEEESTD.2016.7740858). She is a founding member (the only Italian researcher at the constitutive act together with Prof. Alessandro Sperduti of the University of Padua) of the IEEE task force on Process Mining, appointed as an expert on research topics on Process Mining.

He is among the authors of the Process Mining Manifesto, a document drawn up by 75 international process mining experts to define the guidelines and challenges of the near future, for developers, researchers, consultants and users who deal with Process Mining and Business Intelligence. The manifesto has been translated into 16 languages. It is one of the 5 authors of the Italian version.

### 10.1 AWARDS

His research activities have received the following awards:

- (1) The work "Coalition Formation with Logic-Based Agents" [22], in which techniques for reasoning about coalition formation are described, has been judged as one of the best contributions accepted in the

2016 at the "International Conference on Multi-Agent Systems and Agreement Technologies (EUMAS/AT)".

An extended version was then invited to appear in the journal AI Communications [20].

- (2) The work [47, ICEIS'08] about the use of classification techniques in process management contexts in 2008 was awarded the Best Paper Award (area "Artificial Intelligence and Decision Support Systems") in the tenth edition of the "International Conference on Enterprise Information Systems (ICEIS)".
- (3) The work [48, DAWAK'06] about the use of co-clustering methods for process mining was judged as one of the best contributions accepted in 2006 at the "International Conference on Data Warehousing and Knowledge Discovery (DAWAK)". An extended version was then invited to a special issue of the International Journal of Datawarehousing and Mining [34, JDWM'07].

## 10.2 DISSEMINATION OF RESULTS

His works have received about 3634 citations, with h-index=20 (Google Scholar, February 2024).

For several consecutive months (from January to March 2014), his work on inverse frequent itemset min-ing was ranked in February 2014, in the fourth place among the top ten "**Top 10 Downloaded Articles (past 6 weeks)**" of the prestigious journal ACM Transactions on Knowledge Discovery from Data (TKDD)[27] (<http://dl.acm.org/pub.cfm?id=J1054>).

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His work "Process Mining Manifesto" [68] has obtained 1805 citations and is considered the milestone of Process Mining (<https://www.tf-ym.org/resources/manifesto>). The document has been translated into 16 languages, including Italian, of which he edited and wrote the translation together with 4 other co-authors (<https://www.tf-ym.org/resources/manifesto>).

## 10.3 SPEAKER AT CONGRESSES/CONFERENCES

He has reported on the results of his research activities at national and international conferences and workshops, He also held the following invited seminars:

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19-7-2023 | Invited Talk in the FIRST-YEAR TEACHING WORKSHOP ON "BDLT: STATE OF THE ART AND APPLICATIONS", San Michele complex in Fano, 17-21, July 2023. "Title of the talk: Blockchain for Industry 4.0: Emergence, Challenges and Opportunities"                                                                                                                                                                                                                       |
| 25-3-2023 | Invited Talk for the ROBOAI-CON2023 conference, 23-25 March, Rome, Italy. Title: Fed-erated learning in Radiomics applications: Challenges and Emerging Applications.                                                                                                                                                                                                                                                                                         |
| 20-3-2023 | Intervention Scheduled as part of the series of seminars on Artificial Intelligence for Medicine, organized by the Department of Mathematics and Computer Science whose main objective is to create opportunities for discussion and debate between Artificial Intelligence researchers and those of the Medical Area for the start of interdisciplinary research. Title of the Intervention: Open Network for RADIOmicA/radiogenoMica CoperativA- RADIOAMICA |
| 16-6-2023 | Opening seminar of the Advanced Management Training Course for Healthcare Managers, established by the Department of Business and Legal Sciences of the University of Calabria, entitled "Introduction to Digital Innovation for Health".                                                                                                                                                                                                                     |
| 8-2-2023  | Invited seminar "Federated Learning and applications in radiomics" within the cycle of thematic seminars organized by the interdepartmental research center STITCH - Sapienza information-based Technology Innovation Center for Health, La SAPIENZA, University of Rome                                                                                                                                                                                      |

|            |                                                                                                                                                                                                                                                                                                                                           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18-6-2022  | Invited seminar at the conference on "Artificial Intelligence and Telemedicine for Monitoring and Advanced Diagnostics" established by the Order of Physicians-Surgeons and Dentists of CS, event accredited by the Ministry of Health (10 credits). Title: Federated Learning and its applications to Radiomics.                         |
| 16-12-2021 | Invited paper at 3rd Distributed Ledger Technology Working Group Workshop on Multidisciplinary Aspects, organized by the Italian DLT group at SellaLab - Course Giuseppe Garibaldi, 203, Salerno. Title: Distributed Ledgers for Sharing Economy and Digital Rights.                                                                      |
| 4/30/2015  | "Process mining meets security" within the module "Management of Research Projects" Industrial and Development", at the headquarters of the Cyber Security District.                                                                                                                                                                      |
| 6/11/2005  | He presented his research activities in the field of workflow mining at the "Infor-mation Systems (IS)" group of the Technical University of Eindhoven. The invitation was made by Prof. Will van Der Aalst, coordinator of the group and absolute point of reference in research on networks Petri dishes and Process Mining techniques. |

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## 11 EDUCATIONAL ACTIVITIES

He has carried out teaching activities continuously since the Academic Year 2000/2001 at the University of Calabria; furthermore, from the academic year 2000/2001 to the academic year 2006/2007 it has continuously carried out activities teaching at the Magna Grecia University of Catanzaro. Overall, in 22 academic years, he held the ownership of teaching 37 courses for a total of 215.5 CFU, and 18 exercise assignments for a total of 93 CFU.

The following section contains details of the teaching activity carried out. For all teaching assignments reported below was President of the relevant commissions established for the profit exams, while for all the exercise assignments she was a member of the relevant commissions. Furthermore, in the degree courses in Engineering Management (Bachelor's and Master's) and in the Computer Engineering degree course (Master's) at the University of Calabria has been a member of over 50 commissions established for degree exams.

### 11.1 TEACHING ACTIVITIES IN DEGREE COURSES

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Academic Year 2023/2024 | <p>Teacher of the course "Programming Techniques" (6CFU), Three-year Degree Course in Management Engineering at the University of Calabria.</p> <p>Teacher of the exam commission for the course "Computer methods for process analysis" (6CFU), Master's Degree in Management Engineering at the University of Calabria. The course has been moved to the second year of the master's degree and therefore in 2024 the following courses have been only exams delivered.</p> <p>Teacher of the course "Fundamentals of Computer Science" (9CFU), Bachelor's Degree Course in Management Engineering at the University of Calabria.</p> |
| Academic Year 2022/2023 | <p>Teacher of the course "Computer methods for process analysis" (6CFU), Degree Course Master's Degree in Management Engineering at the University of Calabria.</p> <p>Teacher of the course "Fundamentals of Computer Science" (9CFU), Bachelor's Degree Course in Management Engineering at the University of Calabria.</p>                                                                                                                                                                                                                                                                                                           |
| Academic Year 2021/2022 | <p>Teacher of the course "Computer methods for process analysis" (6CFU), Degree Course Master's Degree in Management Engineering at the University of Calabria.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | Teacher of the course "Fundamentals of Computer Science" (9CFU), Bachelor's Degree Course in Management Engineering at the University of Calabria                                                                                                                                                                                                                                                                                                              |
| Academic Year 2020/2021 | <p>Lecturer of the course "Computer methods for process analysis" (6CFU), Master's Degree in Management Engineering at the University of Calabria.</p> <p>Professor of the course "Fundamentals of Computer Science" (6CFU), Bachelor's Degree in Management Engineering at the University of Calabria Professor of the course</p> <p>"Fundamentals of Computer Science" (6CFU), Bachelor's Degree in Electronic Engineering at the University of Calabria</p> |
| Academic Year 2019/2020 | <p>Lecturer of the course "Computer methods for process analysis" (6CFU), Master's Degree in Management Engineering at the University of Calabria.</p> <p>Professor of the course "Fundamentals of Computer Science" (6CFU), Bachelor's Degree in Management Engineering at the University of Calabria Professor of the course</p> <p>"Computer Science for Social Sciences" (6CFU), Master's Degree in Political Science at the University of Calabria</p>    |
| Academic Year 2018/2019 | <p>Lecturer of the course "Computer methods for process analysis" (6CFU), Master's Degree in Management Engineering at the University of Calabria.</p> <p>Teacher of the course "Fundamentals of Computer Science" (6CFU), Bachelor's Degree Course in Management Engineering at the University of Calabria</p> <p>Teacher of the course "Basic Computer Science Laboratory" (3CFU), Three-year Degree Course in Tourism Sciences, University of Calabria</p>  |
| Academic Year 2017/2018 | <p>Teacher of the course "Computer methods for process analysis" (6CFU), Master's Degree in Management Engineering at the University of Calabria.</p> <p>Professor of the course "Fundamentals of Computer Science" (6CFU), Bachelor's Degree in Management Engineering at the University of Calabria Professor of the course</p> <p>"Basic Computer Science Laboratory" (3CFU), Bachelor's Degree in Tourism Sciences at the University of Calabria</p>       |
| Academic Year 2016/2017 | <p>Lecturer of the course "Computer methods for process analysis" (6CFU), Master's Degree in Management Engineering at the University of Calabria.</p> <p>Teacher of the course "Fundamentals of Computer Science" (6CFU), Bachelor's Degree in Management Engineering at the University of Calabria</p> <p>Teacher of the course "Basic Computer Science Laboratory" (3CFU), Three-year Degree Course in Tourism Sciences, University of Calabria</p>         |
| Academic Year 2015/2016 | <p>Lecturer of the course "Computer methods for process analysis" (6CFU), Master's Degree in Management Engineering at the University of Calabria.</p> <p>Teacher of the course "Laboratory of algorithms and data management" (6CFU), Degree Course in Computer Engineering at the University of Calabria.</p>                                                                                                                                                |
| Academic Year 2014/2015 | Teacher of the course "Programming Techniques" (9CFU), Degree Course in Management Engineering at the University of Calabria.                                                                                                                                                                                                                                                                                                                                  |
| Academic Year 2013/2014 | Teacher of the course "Programming Techniques" (9CFU), Degree Course in Management Engineering at the University of Calabria.                                                                                                                                                                                                                                                                                                                                  |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Academic Year 2012/2013 | Teacher of the course "Programming Techniques" (6CFU), Degree Course in Engineering Management at the University of Calabria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Academic Year 2011/2012 | Teacher of the course "Programming Techniques" (6CFU), Degree Course in Engineering Management at the University of Calabria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Academic Year 2009/2010 | Teacher of the course "Object Oriented Programming" (9CFU), Degree Course in Management Engineering at the University of Calabria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| AA 2008/2009            | Teacher of the course "Object Oriented Programming" (9CFU), Degree Course in Management Engineering at the University of Calabria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| AA 2007/2008            | Teacher and course of "Introduction to Computer Science" (2CFU), Degree Course in Engineering Management at the University of Calabria.<br><br>Teacher of the course "Object Oriented Programming" (5CFU), Degree Course in Management Engineering at the University of Calabria.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| AA 2006/2007            | Teacher and lecturer of the course "Software Engineering" (4.5CFU), Degree Course in Computer Science and Biomedicine, at the Magna Graecia University of Catanzaro.<br><br>Teacher of the course "Fundamentals of Computer Science" (5CFU), Degree Course in Computer Science and Biomedicine, at the Magna Graecia University of Catanzaro.<br><br>Teacher and lecturer of the course "Data Mining and Knowledge Discovery" (6CFU), Master's Degree in Health and Biomedical Systems Engineering, at the Magna Graecia University of Catanzaro.<br><br>Teacher and course of "Introduction to Computer Science (Course B)" (2CFU), Degree Course in Management Engineering at the University of Calabria. |
| AA 2005/2006            | Teacher and lecturer of the course "Software Engineering" (4.5CFU), Degree Course in Computer Science and Biomedicine, at the Magna Graecia University of Catanzaro.<br><br>Teacher of the course "Fundamentals of Computer Science" (5CFU), Degree Course in Computer Science and Biomedicine, at the Magna Graecia University of Catanzaro.<br><br>Teacher and lecturer of the course "Data Mining and Knowledge Discovery" (6CFU), Master's Degree in Health and Biomedical Systems Engineering, at the Magna Graecia University of Catanzaro.<br><br>Teaching assistant of the course "Languages and Formal Systems" (6CFU), Master's Degree in Computer Engineering, at the University of Calabria.    |
| AA 2004/2005            | Teacher of the Software Engineering course (4.5 Training Credits). Degree Course in Computer and Biomedical Engineering, at the Magna Graecia University of Catanzaro.<br><br>Teaching assistant of the course "Languages and Formal Systems" (6CFU), Master's Degree in Computer Engineering, at the University of Calabria.<br><br>Tutoring assignment for the course Fundamentals of Computer Science I (teacher Prof. Saccà). Course of Computer and Biomedical Engineering Degree, at the Magna Graecia University of Catanzaro.                                                                                                                                                                       |
| AA 2003/2004            | Tutor of the course of Fundamentals of Computer Science (teacher Prof. G.Manco), Course of Degree in Engineering.<br><br>Teaching assistant of the course Introduction to Computer Science (teacher Prof. S.Basta), Degree Course in Engineering.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|              |                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Teaching assistant of the course of Formal Languages and Systems (teacher Prof. D.Saccà), Master's Degree in Computer Engineering.                                                             |
|              | Teaching assistant of the course Fundamentals of Computer Science I (teacher Prof. Saccà). Degree course in Computer and Biomedical Engineering, at the Magna Graecia University of Catanzaro. |
| AA 2002/2003 | Teaching assistant of the course Fundamentals of Computer Science (teacher Prof. G.Manco), Degree Course in Engineering.                                                                       |
|              | Lecturer of the Introduction to Computer Science course (teacher Prof. G.Manco), Engineering Degree Course.                                                                                    |
|              | Lecturer of the course Introduction to Computer Science (teacher Prof. C.Mastroianni), Degree Course in Engineering.                                                                           |
|              | Tutoring assignment for the course Fundamentals of Computer Science I. Degree Course in Computer and Biomedical Engineering, at the Magna Graecia University of Catanzaro.                     |
|              | Teaching assistant of the Software Engineering course, (teacher Prof. P. Veltri). Degree course in Computer and Biomedical Engineering, at the Magna Graecia University of Catanzaro.          |
| AA 2001/2002 | Teaching assistant of the course Introduction to Computer Science (teacher Prof. S. Flesca), Degree Course in Computer Engineering.                                                            |
|              | Lecturer of the Computer Science Laboratory course, (teacher Prof. L. Palopoli). Faculty of Agriculture, University of Reggio Calabria.                                                        |
|              | Tutoring assignment for the course Fundamentals of Computer Science I. Degree Course in Computer and Biomedical Engineering, at the Magna Graecia University of Catanzaro.                     |
| AA 2000/2001 | Teaching assistant of the course Fundamentals of Computer Science (teacher Prof. G. Manco), Degree Course in Engineering.                                                                      |

## 11.2 TEACHING ACTIVITIES FOR THE DOCTORATE

Professor of the PhD course in ICT at the University of Calabria: Binary Analysis with Applications to Machine and Deep Learning, 12:00 p.m., 3 CFU, academic year 2023-2024.

Teacher of the PhD course: Binary Analysis with Applications to Machine and Deep Learning, 12pm, 3 CFU, academic year 2022-2023

Tutorial on "ProM - the Process Mining framework" - lectures of the "Doctoral School in Computer Engineering", organized by the Computer Engineering Group (GII), 29-9-2010.

## 11.3 TEACHING ACTIVITIES AT MASTERS AND/OR TRAINING COURSES

|                                      |                                                                                                                                                                                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| December 2023                        | TEACHING Assignment – Master's Degree in Digital Document Conservator – PERSEO academic year 2022-2023. Module 2 - Methodologies and techniques for the conservation of digital materials - Submodule 2.3 - Information classification and coding systems.    |
| from September 2023 to October 2023: | He held a teaching position for the "Advanced Management Training Course for Health Managers", at the Department of Legal and Business Sciences DISCAG of UNI-CAL. The topics covered in the lessons held concerned the theme "Digital innovation for health" |
| from May 2020 to June 2020:          |                                                                                                                                                                                                                                                               |

|                                      |                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | He held a teaching position in the "Workflow and Processes" module at the post-graduate master's degree in IoT held at the DIMES department of UNICAL.                                                                                                                                                                      |
| from May 2019 to June 2019:          | He held a teaching position in the "Workflow and Processes" module at the post-graduate master's degree in IoT held at the DIMES department of UNICAL.                                                                                                                                                                      |
| from May 2015 to June 2015:          | He held a teaching position in the "Workflow and Processes" module at the post-graduate master's degree Ba2Know held at the mathematics department of UNICAL.                                                                                                                                                               |
| since October 2012 in November 2012: | He held a teaching position in the "Workflow and Processes" module at the post-graduate master's degree TETRIS held at ICT Sud.<br>He held a teaching position in the "Workflow and Processes" module at the post-graduate master's degree FRAMER held at the mathematics department of UNICAL.                             |
| from April 2010 to June 2010:        | He held a teaching position in the "Workflow and Processes" module at the Open-Knowtech laboratory of ICT Sud.                                                                                                                                                                                                              |
| from 4/7/2005 to 20/7/2005:          | He held a teaching position in the modules "Workflow Systems" and "Information Systems" at the post-graduate master's degree MENTE held at the University of Calabria in collaboration with "Sviluppo Italia".                                                                                                              |
| from 26/4/2005 to 26/5/2005:         | He held a teaching position in the modules "Business Information Systems" and "Software solutions design for the web", within the IFTS - POR 3.7a course Code 67 Year 2002/2003 organized by the IPSIA G. Ferraris institute - CATANZARO.                                                                                   |
| from 4/2/2005 to 18/3/2005:          | He held a teaching position in the UFC "Introduction to Algorithms" and "Programming Languages and Methods", within the IFTS course "Higher Technician for Software Development" organized at the S. Maria Ausiliatrice Institute of Soverato in collaboration with the University of Calabria and Golem Software of Palmi. |
| from 24/2/2004 to 3/3/2004:          | He held a teaching position for the modules of "Knowledge Management Architectures", within the "Master in Knowledge Management" organized by the Center for Economic and Social Engineering (CIES) of Rende (CS).                                                                                                          |
| from 10/5/2002 to 10/19/2002:        | He has held a teaching position for the modules (i) "Introduction to Programming" and (ii) "Information processing systems", as part of the professional training course in Graphic and Web Designer organized by the ISIM-Srl company of Vibo Valentia (VV).                                                               |

#### 11.4 SEMINAR-TYPE ACTIVITIES, EXERCISES AND TUTORING

In addition to the tutorial duties listed in section 11.1, she has held numerous seminars and has carried out an intense tutoring activity for the Bachelor's and Master's degree students in Management Engineering. In particular, since the academic year 2008, she has been responsible for the supervision, every year, of about 20 students enrolled in the first year, according to a "freshmen" support plan provided for in the Management Engineering degree course; In this course, she has also supervised as supervisor over 50 degree theses.

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## 12 ACTIVITIES ON DOCTORATES AND POST-DOC COURSES

- Since 2007, he has been a member of the Teaching Staff of the PhD in Systems Engineering and Computer Science at the DIMES Department of the University of Calabria, supervising 3 PhD students on Blockchain and Deep/Machine Learning topics.
- From 2021 to July 2023 she was president of the evaluation commission for admission to the years following the first and the final exam, for the ING-INF/05 sector within the above-mentioned doctorate; She was also a member of the examination commission for admission to the 38th cycle by rectoral decree DR464/2023.
- She was a member of the evaluation commission for the final exam to obtain the PhD title in "Mathematics and Computer Science" - XXXV cycle of the University of Calabria by rectoral decree DR423/2023.
- She was a member of the evaluation commission for the final exam to obtain the title of Doctor of Research in "Mathematics and Computer Science" - XXXIV cycle of the University of Calabria by rectoral decree DR428/2023.
- Since 2022 he has been a member of the National Doctoral College in BLOCKCHAIN AND DISTRIBUTED LEDGER TECHNOLOGY, administrative headquarters of the School of Advanced Studies - International Doctoral School of the University of Camerino.
- She was appointed member of the examination board for admission to the National Doctorate in BLOCKCHAIN AND DISTRIBUTED LEDGER TECHNOLOGY (XXXVIII CYCLE academic year 2022/2023 and XXXIX CYCLE academic year 2023/2024).
- She was appointed as the contact person for the Curriculum Industry 4.0 of the National Doctoral College in BLOCKCHAIN AND DISTRIBUTED LEDGER TECHNOLOGY for the XXXVIII and XXXIX cycles.

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## 13 TECHNOLOGY TRANSFER ACTIVITIES

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EVO-BI               | He is a founding partner and member of the Board of Directors of the company EVO-BI srl, a start up registered in the special section of Innovative Start Ups in the Company Register of COSENZA (CS) at no. 03643580784, dated 04/07/2019. The company EVO-BI srl has been recognized as an academic SPIN OFF of the University of Calabria (resolution of the University Board of Directors dated 4/7/2019). The SPIN OFF aims to use academic skills in the field of Big Data management and the use of Artificial Intelligence techniques to develop EVO-BI, a platform with Business Analytics and Business Intelligence functionalities for the management control of SMEs.                                          |
| OPEN SOURCE SOFTWARE | He is among the developers of the ProM platform ( <a href="http://www.processmining.org/people/dev">http://www.processmining.org/people/dev</a> ), the reference suite for the process mining community that implements process mining techniques in a Plug-In environment. 22 international commercial partners and academic institutions, as well as two Dutch spin-offs (futura and fluxigon), are involved in the development of ProM.<br>Part of the results of his research have been implemented in two plug-ins for this platform [32,35]: "Process Discovery" plug-in for extracting a DWS process model (see Sect. 4.1) and "Analysis" plug-in that adds analysis functionality based on process log clustering. |

Execution



|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OpenKnowTech | Part of the results of her research in [33,36] have been used in the prototyping of a process mining suite (PMS) on behalf of the company "Exeura Srl", a spin-off of the University of Calabria operating in the ICT sector. The suite has been enriched with process mining functionalities [32] within the activities of the public-private laboratory "OpenKnowTech" on the implementation objective "Open Source for process management and Business Process Intelligence", of which she was a member for the research unit of the University of Calabria. |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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## 14 MEMBER OF JUDGING COMMITTEES FOR PUBLIC COMPETITIONS

She has been appointed as a member of the following committees for the following entities:

|                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ICAR-CNR                                                                                                                                      | <p>Examination committee for the selection procedure based on qualifications and interview pursuant to art. 8 of the "Regulations concerning the hiring of personnel with fixed-term employment contracts", for the hiring, pursuant to art. 83 of the CCNL of the "Education and Research" Sector 2016-2018, signed on 19 April 2018, of 3 (three) personnel with the professional profile of researcher - level III, at the Institute for High Performance Computing and Networks of the National Research Council - Rende (CS) - Announcement no. 400.10 ICAR - PNRR SERICS Project.description.</p> <p>Judging commissions for additional public selections for admission to doctoral courses - XXXVIII Cycle AA 2022/2023 with the assignment of doctoral scholarships financed by NextGenerationEU as part of the National Recovery and Resilience Plan (PNRR), financed by INPS and financed as part of the "Municipal Doctorates" initiative</p> |
| University of Bari Aldo Moro RTDA Selection - code PNRR_PE_06                                                                                 | Public selection for the coverage of 52 positions of University Researcher with a fixed term, pursuant to art.24, co.3, lett. a), l.240/10, within the PNRR, Mission 4-Component 2-Investment 1.3 – Funded by the EU – NextGenerationEU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| University of Verona Evaluation procedure for 1 position of Associate Professor                                                               | ssd ING-INF/05 Information Processing Systems Code 2022paps24003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| University of Verona Selection procedure for 1 position of associate professor                                                                | ssd ING-INF/05 - Extraordinary Plan Code 2021paps18001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| University of Pisa: Commission RIC2021PON_A35 - Department of Information Engineering - SSD ING-INF/05 - Information Processing Systems       | Selection procedures for the hiring of 43 fixed-term researcher positions, pursuant to art. 24, paragraph 3, letter a) L. 240/2010 (PON 2014-2020 ACTIONS IV.4 AND IV.6)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| University of Calabria: Procedure Commission R52_2023_ING-INF/05                                                                              | Selection notice 16 RTD A - PNRR – Recruitment of fixed-term researchers pursuant to article 24, paragraph 3, letter a), Ln240/2010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| University of Calabria: Procedure Commission R49_2023_ING-INF/05                                                                              | Selection notice 16 RTD A - PNRR – Recruitment of fixed-term researchers pursuant to article 24, paragraph 3, letter a), Ln240/2010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| University of Calabria: Public competition, based on qualifications and examinations, for the recruitment of one unit of category D personnel | - economic position D1 - in the technical, technical-scientific and data processing area, with an indefinite contract, reserved exclusively for the subjects referred to in art. 1 of Law 12 March 1999, n. 68, Professional Profile of Computer Science-Application for the needs of the Administration and Central Services.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

DIMES-University of Calabria: Public competition 8 units category c - DD n° 173 of 03/05/2022 profile TEC\_DIMES\_C Public competition, for qualifications and exams, for the recruitment of 8 research laboratory technicians, to be classified in category C, economic position C1, technical area, technical-scientific and data processing, permanent

University of Calabria: 1062\_R28\_INNOVATION\_09H1\_ING\_INF05\_DIMES Selection announcement no. 45  
RTD-A (DM 1062 of 10 August 2021)

University of Reggio Calabria: Selection Committee for the selection of a fixed-term Researcher, for the Competition Sector 09/H1 SSD ING-INF/05 at the Department of Information, Infrastructure and Sustainable Energy Engineering (DIIES), pursuant to the PON "Research and Innovation" pursuant to Ministerial Decree 1062/2021. Call DR n° 260 of 19/10/2021 (notice published in the Official Journal n° 82 of 15/10/2021)

University of Calabria: Member of the judging commission of a public competition by examination, for the recruitment of a category C staff unit to be assigned to the ICT Center of the University of Calabria (directorial decree no. 739 of 15 05 2020). .

University of Calabria: Member of the Judging Committee in the selection of candidates for the "Cyberchallenge program", for the local unit of the University of Calabria and on behalf of CINI in both the 2019 and 2020 editions.

Conservatory of Music: Internal member of the judging commission of a public competition by examination, for the creation of a ranking of suitable candidates for the professional profile of Collaborator - Area III - CCNL of the Afam Sector, at the Stanislao Giacomoantonio Conservatory of Music of Cosenza (presidential decree n. 8052 of 20.10.2015).

University of Calabria: Member of the judging commission of a public competition by examination, for the recruitment of three category C personnel to be assigned to the strategic unit of the university statistics service and decision support of the University of Calabria (directorial decree no. 898 of 14.05.2014).

University of Calabria: President of the Selection Committee in the public selection procedure, based on qualifications and interview, for the awarding of 1 research grant relating to the Digital Identity and Service Accountability Research Program (IDService) at the Department of Computer Engineering, Modeling, Electronics and Systems (DIMES) - UNICAL, issued with DD n° 8 of 10/01/2018, pursuant to Law n. 240/2010, articles 18, 22, 24 and 29 and the Regulation for the Awarding of Grants for Collaboration in Research Activities, approved with DR n. 1655 of 08/09/2014.

University of Calabria: Member of the judging commission in the selection of candidates for the "2nd LEVEL MASTER in INFORMATION SECURITY" and the "Master in Ethical Hacking", issued with DD nr. 217 of 11/30/2017.

University of Calabria: Member of the judging committee in the selection of 54 graduates to receive training actions of the 3 Training Projects "PON03PE\_00032\_1", "PON03PE\_00032\_2" and "PON03PE\_00032\_3" within the Cybersecurity project, enacted with DD nr. 377 of 22/10/2014.

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## 15 OTHER INSTITUTIONAL ACTIVITIES AT THE UNIVERSITY OF CALABRIA

- He is a member of the university working group for the drafting of the guidelines for the accreditation of doctorates

- Since 2007 he has been a member of the Management Engineering Degree Course at the University of Calabria.
- Since 2015 he has been a member of the commission established by the DIMEG Department for the assignment of academic tutors in SSDING ȳ INF/05.
- She is a member of the commission for the assignment of teaching positions in ING-INF/05 for the degree course in Management Engineering and Computer Engineering, for which she also held the role of President; She was President of the Commissions for tenders for hardware supplies of the Department.
- She was President of the commission for the election called on March 25, 2021, of a representation of the holders of research grants within the Council of the Department of Computer Engineering, Modeling, Electronics and Systems - DIMES.
- Since 2019 he has been a member of the "Didactics" commission for the Management Engineering Degree Course;
- She was President of the commission for the election of the professors' representatives called on 16-6-2021  
Full professors, associate professors, researchers and technical-administrative staff within the Board of the Department of Computer Engineering, Modeling, Electronics and Systems - DIMES
- She has been a member of numerous committees for the awarding of research grants and collaboration contracts.  
ration at the University of Calabria.
- She was a proposing member and member of the scientific council of the MASTER II LEVEL INFORMATION SECURITY,  
approved with DRn902 of 13/07/2017.

He also held the following positions:

from 2014 to 2019 With DR n.1191 of 26/6/2014 and with DR n.974 of 29/07/2016 he held the role and functions of President of the "Direction and Control Commission" established to carry out direction and control functions of the activities of the childhood hub service of the University of Calabria.

from 2013 to 2019 With DR n.2599 of 10/12/2013 she covered the role and functions of "Rector's Delegate for the coordination and supervision of the correct performance of nursery school activities and with DR n.1687 of 10/09/2014 of the service of the Childhood Center of the University of Calabria.

2009 Associate member for the Information Engineering sector of the First and Second Session 2009 of the State Exams for the qualification to practice the professions of Engineer (old system and new system section A) and Junior Engineer (new system section B);

2009 Responsible for the management of the university orientation event "Campus Live" (2009 edition) for the skills of the Degree Course in Computer Engineering.

2008 Contact person for the Degree Course in Computer Engineering and the Specialist Degree Course in Computer Engineering, for the organization of the 2008 Campus Lecture days.

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## 16 LIST OF MAIN PUBLICATIONS

For the complete list, refer to the link on scholar \$[https://scholar.google.com/citations?hl=it&user=N1vtMtQAAAAJ&view\\_op=list\\_works&sortby=pubdate](https://scholar.google.com/citations?hl=it&user=N1vtMtQAAAAJ&view_op=list_works&sortby=pubdate)\$scholar page and the link on scopus <https://www.scopus.com/authid/detail.uri?authorId=6701539606>

and the link on scopus <https://www.scopus.com/authid/detail.uri?authorId=6701539606>

- 1 D Thakur, A Guzzo, G Fortino,  
"Hardware-algorithm co-design of Energy Efficient Federated Learning in Quantized Neural Network"  
Internet of Things, 26,  
101223 (2024)
  
- 2 M Talia et al.  
"The G Protein Estrogen Receptor (GPER) is involved in the resistance to the CDK4/6 inhibitor palbociclib in breast cancer"  
  
Journal of Experimental & Clinical Cancer Research 43 (1), 171  
(2024)
  
- 3 D Thakur, A Guzzo, G Fortino.  
"A Feature Fusion Method Integrating Gradient Boosted Feature Selection and Deep Learning for Performance-Aware Physical Activity Recognition"  
HUMAN-CENTRIC COMPUTING AND INFORMATION SCIENCES 14(2024)
  
- 4 Pian Qi, D. Chiaro, A. Guzzo, M. Ianni, G. Fortino, F. Piccialli, "Model aggregation techniques in federated learning: A comprehensive survey."  
Future Gen. Computing Syst., 150: 272-293 (2024)
  
- 5 D Thakur, A Guzzo, G Fortino,  
"Energy Aware Federated Learning with Application of Activity Recognition"  
The 21st IEEE International Conference on Pervasive Intelligence and Computing (PICom 2023), 2023
  
- 6 C Adornetto, A Guzzo, A Vasile,  
"Automatic Medical Report Generation via Latent Space Conditioning and Transformers"  
The 21st IEEE International Conference on Dependable, Autonomic & Secure Computing (DASC 2023), 2023
  
- 7 G. Fortino, C. Greco, A. Guzzo, M. Ianni, "Identification and prediction of attacks to industrial control systems using temporal point processes"  
Journal of Ambient Intelligence and Humanized Computing, 14 (5), 2023
  
- 8 D. Thakur, A. Guzzo, G. Fortino, "Attention-based Multihead Deep Learning Framework for online activity monitoring with Smartwatch Sensors"  
IEEE Internet of Things, IEEE, 2023
  
- 9 S. Guo, H. Zhang, Y. Gao, H. Wang, L. Xu, Z. Gao, A. Guzzo, G. Fortino, "Survival prediction of heart failure patients using motion-based analysis method"  
Computer Methods and Programs in Biomedicine Elsevier, , 2023
  
- 10 A. Ali, M. Pasha, A. Guerrieri, A. Guzzo, Internet of Medical Things trial"  
  
IEEE Transactions on Network Science and Engineering, IEEE, 2023

- 11** J. Chen, H. Xiao, Y. Zheng, MM. Hassan, M. Ianni, A. Guzzo, G. Fortino,  
"DKSM: A Decentralized Kerberos Secure Service-Management Protocol for Internet of Things"  
Internet of Things,  
Elsevier, Netherlands, 2023
- 12** A. Guzzo, G. Fortino, G. Greco, M. Maggiolini,  
"Data and model aggregation for radiomics applications: Emerging trend and open challenges"  
Information Fusion, 13 July 2023  
Elsevier, Netherlands.
- 13** C Felicetti, A Guzzo, G Manco, F Pasqua, E Ritacco, A Roller, D Saccà,  
"Deep Learning/PUF-based Item Identification for Supply Chain Management in a Distributed Ledger Framework"  
Fifth International Conference on Blockchain Computing and Applications 2023
- 14** S. Guo, X. Liu, H. Zhang, Q. Lin, L. Xu, C. Shi, Z. Gao, A. Guzzo, G. Fortino,  
"Causal knowledge fusion for 3D cross-modality cardiac image segmentation"  
Information Fusion,  
Elsevier, Netherlands, 2023
- 15** A. Guzzo, A. Rullo, E. Vocaturo,  
"Process mining applications in the healthcare domain: A comprehensive review"  
Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, 12  
(2), 2022
- 16** G. Fortino, A. Guzzo, M. Ianni, F. Leotta, M. Mecella,  
"Predicting activities of daily living via temporal point processes: Approaches and experimental results"  
Computers & Electrical Engineering, 96,  
2021.
- 17** A. Guzzo, M. Joaristi, A. Rullo, E. Serra, "A  
multi-perspective approach for the analysis of complex business processes behavior"  
Expert Systems with Applications,  
177, 2021
- 18** A. Guzzo, M. Ianni, A. Pugliese, D. Saccà,  
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Future Generation Computer Systems,  
113, 2020
- 19** V. Fionda, A. Guzzo,  
"Control-Flow Modeling with Declare: Behavioral Properties, Computational Complexity, and Tools"  
IEEE Transactions on Knowledge and Data Engineering, 32  
(5), 898-911, 2019
- 20** M. Alviano, G. Greco, A. Guzzo,  
"Coalition formation in social environments with logic-based agents"  
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- 21** G. Greco, A. Guzzo,  
"Constrained coalition formation on valuation structures: Formal framework, applications, and islands of tractability"  
Artificial Intelligence,  
249, 19-46 2017

- 22** G. Greco and A. Guzzo,  
 “Coalition Formation with Logic-Based Agents”, in  
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 Valencia, Spain, December 15-16, 2016, pp. 455-469, LNCS 10207, Springer, ISBN 978-3-319-59293-0.
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 “Data storage and representation”  
 Encyclopedia of Bioinformatics and Computational Biology, 31 August 2018  
 Editor-in-Chiefs: Shoba Ranganathan Kenta Nakai Christian Schonbach Book ISBN: 9780128114148  
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 “Constrained coalition formation on valuation structures: Formal framework, applications and islands of tractability”  
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- 25** A. Guzzo, A. Pugliese, A. Rullo, D. Saccà, and A. Piccolo,  
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 “Process Discovery under Precedence Constraints”,  
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- 27** A. Guzzo, L. Moccia, D. Saccà, and E. Serra,  
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- 39** M. Cannataro, A. Guzzo, and A. Pugliese,  
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Rende, 9 August 2024

Antonella Guzzo